JOHNS HOPKINS
WHITING SCHOOL of ENGINEERING

Introduction to Generative AI

Review of Machine Learning

Outline

- Module Learning Objectives
- Core Probability Concepts in Machine Learning
- Statistics in Machine Learning
- Linear Algebra
- Genetic Algorithms
- Markov Chains

Module Learning Objectives

- Grasp essential mathematical and statistical principles
  - Explain core concepts including probability distributions, expectation, and variance
  - Apply Bayes theorem and understand hypothesis testing
- Apply linear algebra in machine learning
  - Understand vectors, matrices, eigenvalues, and eigenvectors in data manipulation
  - Use matrix decompositions in the implementation of machine learning algorithms
- Learn classical machine learning models
  - Explain Autoregressive (AR) algorithms and their role in time series forecasting
- Develop problem-solving skills using practical examples
  - Implement machine learning algorithms using practical datasets
  - Analyze the performance of different models on real-world data

Introduction

Probability and Statistics

- Framework for modeling uncertainty and making inference
  - Distributions
  - Expectation
  - Variance
  - Bayes theorem
  - Hypothesis testing
  - Sequence models

Linear Algebra

- Utilization and manipulation of high-dimensional data
  - Vector spaces
  - Vectors
  - Matrices
  - Eigenvalues and Eigenvectors
  - Matrix Decompositions

Probability Theory Fundamentals

- The sample space, S of a random experiment, is the set of all possible outcomes
  - Example, a fair coin toss, the sample space is

  S = {Heads, Tails}

- An event is a subset of the sample space
- An event can consist of one or more outcomes
  - Example, let A be the event of getting a head when tossing a coin, then

  A = {Heads}

Probability Axioms

- Probability of an event is a measure of the **likelihood that the event will occur**
- Probability is quantified by a real number between 0 and 1, P:
  - where 0 indicates impossibility, and 1 indicates certainty
- A discrete probability function is any function $P:\mathbb{F}\rightarrow[0,1]$ satisfying the following

Axiom 1 Non-negativity	Axiom 2 Normalization	Axiom 3 Additivity
For any event A $P(A) \geq 0$	Probability of the sample space is 1 $P(S) = 1$	For any two mutually exclusive events A & B $P(A \cup B) = P(A) + P(B)$

Example Rolling a Fair Die

- Consider the random experiment of rolling a fair six-sided die
  - The sample space

    S = {1, 2, 3, 4, 5, 6}

  - Let A be the event of rolling an even number

    A = {2, 4, 6}

  - The probability of A

$$\mathrm{P(A)} = \frac{\text{Number of favorable outcomes}}{\text{Total number of outcomes}} = \frac{3}{6} = \frac{1}{2}$$

Conditional Probabilities

- Conditional probability describes the **probability of an event occurring given that another event has already occurred**

$$\mathrm{P}(A|B) = \frac{\mathrm{P}(A \cap B)}{\mathrm{P}(B)}$$

- where
  - P(A∩B) is the probability that both events A and B occur
  - P(B) is the probability that event B occurs
- The probability of an event before we consider our additional knowledge is called the **prior** probability of the event
- The new probability that posterior results from using our additional knowledge is referred to as the **posterior** probability of the event

Example Rolling Another Die

- Consider the random experiment of rolling a fair six-sided die
  - Let A be the event of rolling an even number
    $A = \{2,4,6\}$
  - Let B be the event of rolling a number greater than 3
    $B = \{4,5,6\}$
  - The probability of A is $P(A) = 3/6 = 1/2$
  - The probability of B is $P(B) = 3/6 = 1/2$
  - The probability of both A and B occurring (an even number greater than 3) is
    $P(A \cap B) = 2/6 = 1/3$
  - Therefore, the conditional probability of A given B is

  $P(A \mid B) = P(A \cap B) / P(B) = (1/3) / (1/2) = 2/3$

Bayes Theorem

- Bayes theorem helps update the probability of a hypothesis based on new evidence
- Bayes theorem relates the conditional and marginal probabilities of random events

$$\mathrm{P}(A|B) = \frac{\mathrm{P}(B|A)\mathrm{P}(A)}{\mathrm{P}(B)}$$

- where
  - P(A | B) is the posterior probability
  - P(B | A) is the likelihood
  - P(A) is the prior probability
  - P(B) is the marginal probability

Example Medical Diagnosis

- Consider a medical test for a disease that is 99% accurate
  - Let A be the event that a person has the disease
    $P(A) = 0.01$
- Let B be the event that the test result is positive
  $P(B \mid A) = 0.99$ (True positive rate)
  $P(B \mid \sim A) = 0.01$ (False positive rate)
- The marginal probability P(B) is calculated as
  - $P(B) = P(B \mid A)P(A)+P(B \mid \sim A)P(\sim A) = 0.0198$
- The posterior probability, $P(A \mid B)$, is therefore
  - $P(A \mid B) = P(B \mid A)P(A) / P(B) \approx 0.5$
- If a person tests positive, then there is a 50% chance they have the disease

Random Variables

- A random variable is a function $X:\Omega\rightarrow\mathbb{R}^n$, (for our purposes n=1)
- Rather than having to work with some irregular event space which differs with every problem, a random variable allows us to model the probabilities of numerical values that are related to the event space
  - An abstract stochastic process that generates numbers with a certain probability distribution
  - The word stochastic simply means 'probabilistic' or 'randomly generated,' but is especially commonly used when referring to **a sequence of results** assumed to be generated by some underlying probability distribution
- A discrete random variable is a function $X:\Omega\rightarrow S$, where S is a countable subset of $\mathbb{R}$
- If $X:\Omega\rightarrow\{0,1\}$ then X is called an indicator random variable or a Bernoulli trial

Binomial Distribution

- Models **the number of successes** in a fixed number of independent Bernoulli trials, each with the same probability of success
- The probability mass function (PMF) of the Binomial distribution is given by

$$\mathrm{P}(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$$

- where
  - X is the number of successes
  - n is the number of trials
  - k is the number of successes in those trials
  - p is the probability of success in each trial

Poisson Distribution

- Models **the number of events** occurring in a fixed interval of time or space, given the events occur with a constant mean rate and independently of the time since the last event
- The probability mass function of the Poisson distribution is given by

$$\mathrm{P}(X = k) = \frac{\lambda^k e^{-\lambda}}{k!}$$

- where
  - X is the number of events
  - λ is the average rate of occurrence
  - k is the number of occurrences

Normal (Gaussian) Distribution

- The Normal distribution, also known as **the Gaussian distribution**, is a continuous probability distribution characterized by its bell-shaped curve
- For many (measurable) entities in the world, such as the heights or IQs of people, one gets a distribution that is known in the media as a bell curve
- The probability density function (PDF) of the Normal distribution is given by

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

- where
  - μ is the mean
  - σ^2 is the variance

Exponential Distribution

- Models **the time between events in a Poisson process**
- The probability density function of the Exponential distribution is given by

  $f(x) = \lambda e^{-\lambda x}$

- where
  - λ is the rate parameter

Uniform Distribution

- Models an **equal likelihood of all outcomes** in each interval
- The probability density function of the Uniform distribution is given by

$$f(x) = \frac{1}{b-a} \text{ for } a \leq x \leq b$$

- where
  - a and b are the lower and upper bounds of the interval

Statistical Measures

Skewness

- Quantifies the asymmetry of a probability distribution around its mean
  - Skewed to the left (negative)
  - Skewed to the right (positive)
  - If symmetric, then it is zero

$$\mathbf{S} = \frac{n}{(n-1)(n-2)} \sum_{i=1}^{n} \left(\frac{x_i - \bar{x}}{s} \right)^3$$

- Descriptive analysis, quality control

Kurtosis

- Describes the "tailedness" or the peak of a probability distribution
- How much of the variance is due to outliers as opposed to the overall spread

$$\mathbf{S} = \frac{n(n+1)}{(n-1)(n-2)(n-3)} \sum_{i=1}^{n} \left(\frac{x_i - \bar{x}}{s} \right)^4 - \frac{3(n-1)^2}{(n-2)(n-3)}$$

- Outlier detection, data analysis

Joint Probability Distribution

- The joint probability distribution gives the probability of two (or more) random variables taking on specific values simultaneously
- For two discrete random variables X and Y, the joint probability P distribution is defined as

$$\mathrm{P}(X = x, Y = y)$$

- Consider rolling two fair six-sided dice
  - Let X be the outcome of the first die
  - Let Y be the outcome of the second die

$$\mathrm{P}(X = 3, Y = 5) = \frac{1}{36}$$

Marginal Probability Distribution

- The probability of a single random variable irrespective of the values of other variables
- For discrete random variables X and Y, the marginal probability distribution of X is obtained by summing over all possible values of Y

$$\mathrm{P}(X = x) = \sum_{y} \mathrm{P}(X = x, Y = y)$$

- Roll two fair six-sided dice
  - The marginal probability of X=3

$$= \mathrm{P}(X = 3)$$
$$= \sum_{y=1}^{6} \mathrm{P}(X = 3, Y = y)$$
$$= \sum_{y=1}^{6} \frac{1}{36}$$
$$= \frac{1}{6}$$

Conditional Probability Distribution

- The probability of one random variable given the value of another random variable
- For discrete random variables X and Y, the conditional probability distribution of X given Y is

$$\mathrm{P}(X = x \mid Y = y) = \frac{\mathrm{P}(X=x, Y=y)}{\mathrm{P}(Y=y)}$$

- Drawing two cards sequentially from a standard deck without replacement
  - Let X be the event that the first card is an Ace (symbol ♡ below)
  - Let Y be the event that the second card is an ♡

$\mathrm{P}(X=♡) = 4/52 = 1/13$

$\mathrm{P}(Y=♡ \mid X=♡) = 3/51 = 1/17$

$\mathrm{P}(X=♡, Y=♡) = \mathrm{P}(X=♡)\mathrm{P}(Y=♡ \mid X=♡)$

$= 1/13 * 1/17 = 1/221$

$\mathrm{P}(X\neq♡, Y=♡) = \mathrm{P}(X\neq♡)\mathrm{P}(Y=♡ \mid X\neq♡)$

$= 48/52 * 4/51 = 16/221$

$\mathrm{P}(Y=♡) = 1/221 + 16/221 = 17/221$

Expectation and Variance

- The expectation is the mean or average of a random variable
  - Average or expectation is the **estimate** of the mean
- A measure of the central tendency of its probability distribution

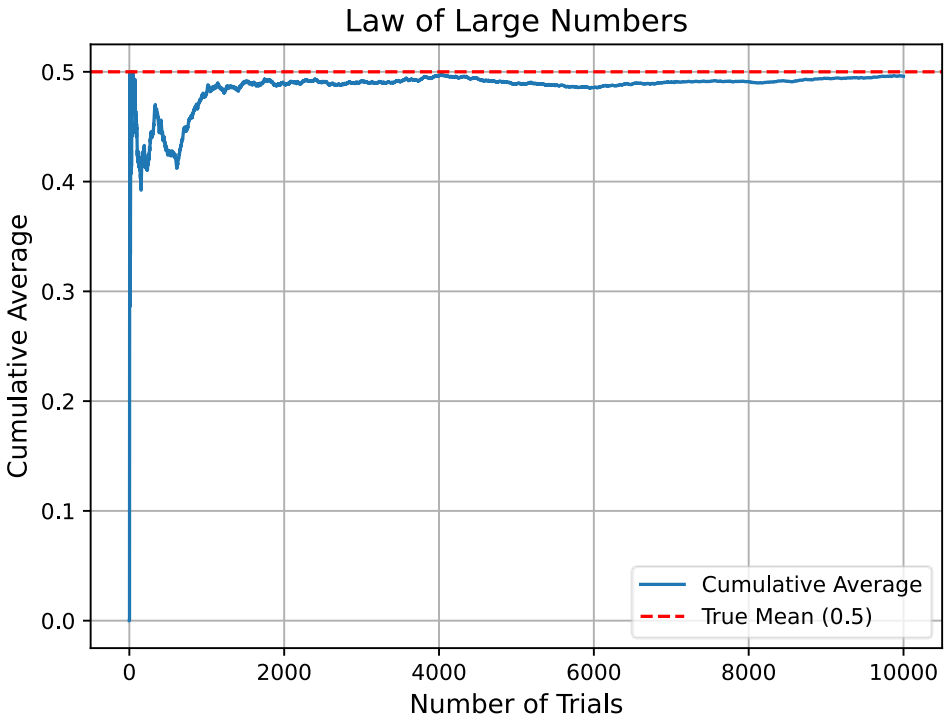
$$\mathbf{E}[X] = \sum_i x_i \mathbf{p}_i$$

- The standard deviation measures the amount of variation or dispersion of a set of values from their mean
- Mean and standard deviation are two major **parameters** of a probability distribution

$$\mathbf{Var}(X) = \mathbf{E}[(X - \mu)^2]$$
$$= \sum_i (x_i - \mu)^2 \mathbf{p}_i$$

The Law of Large Numbers

- As trials increase, the sample average converges to the expected value, making results more predictable
- The Weak law ensures that the sample average converges to the expected value in probability as trials increase
- The Strong law guarantees almost certain convergence of the sample average to the expected value as trials become infinite



The Law of Large Numbers

- Given a sequence of independent and identically distributed (i.i.d.) random variables $X_1, X_2, \ldots, X_n$, each with an expected value (mean) μ and a finite variance σ^2
- Weak Law of Large Numbers (WLLN), for any $\epsilon > 0$,

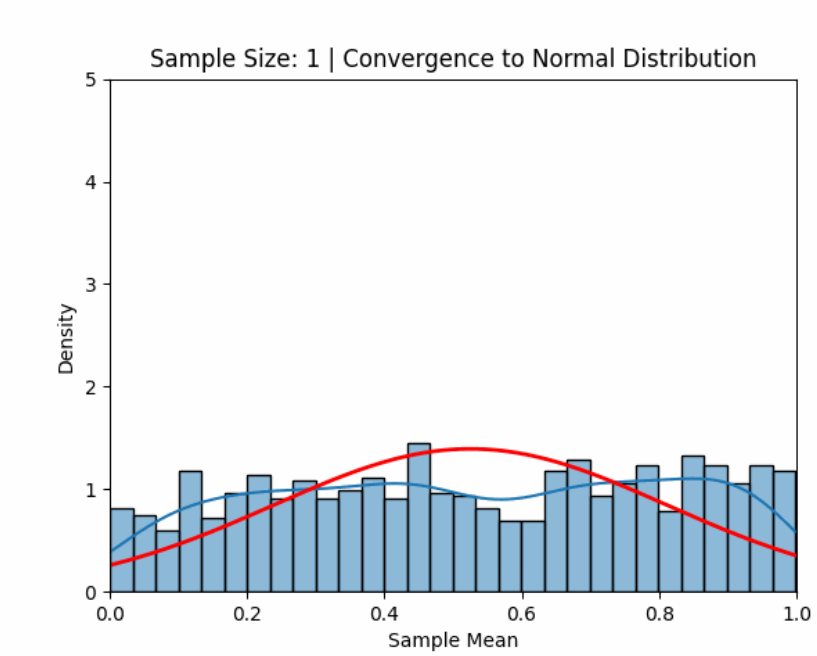
$$\lim_{n\to\infty} \mathrm{P}\left(\left|\frac{1}{n}\sum_{i=1}^{n} X_i - \mu\right| \geq \epsilon\right) = 0$$

- Strong Law of Large Numbers (SLLN)

$$\mathrm{P}\left(\lim_{n\to\infty} \frac{1}{n}\sum_{i=1}^{n} X_i = \mu\right) = 1$$

Central Limit Theorem

- The Central Limit theorem (CLT) states that the sum or average of many independent, identically distributed random variables will form a normal distribution, regardless of the original distribution's shape
- The power of CLT is in its universality is that as sample size increases, the distribution of sample means becomes increasingly normal, enabling reliable predictions and confidence intervals



Hypothesis Testing

- **Null Hypothesis** (H_0) Statement that there is no effect or no difference, serves as the starting assumption for hypothesis testing
- **Alternative Hypothesis** (H_α) Statement that contradicts the null hypothesis, indicating the presence of an effect/difference
- **Type I Error** (α) Occurs when the null hypothesis is rejected when it is actually true
- **Type II Error** (β) Occurs when the null hypothesis is not rejected when it is actually false

Example Hypothesis Testing

- **Scenario** Suspect that the coin is biased (not fair), series of coin flips
- Null Hypothesis (H_0) The coin is fair, meaning the P(Heads) is 0.5
- Alternative Hypothesis (H_1) The coin is not fair, P(Heads) is not 0.5
- **Experiment** Flip the coin 100 times, observe it lands on Heads 60 times
- **Significance Level** (α) Choose a significance level of 0.05 (5%)
  - Accept a 5% chance of rejecting H_0 if it is actually true

- **Test Statistic** The number of flips is large, use a normal approximation to the binomial distribution
  - $z = \frac{(\text{Observed ratio} - \text{Expected ratio})}{\text{Standard Error}}$
  - Observed ratio $\hat{p}$=60/100=0.6
  - Expected ratio p_0=0.5
  - SE = $\sqrt{\frac{p_0(1-p_0)}{n}} = \sqrt{\frac{0.5\times0.5}{100}} = 0.05$
  - $z = \frac{0.6-0.5}{0.05} = 2$ >1.96, **Reject H_0**
- **Decision Rule** Compare z to the critical value of normal distribution
  - Two-tailed α=0.05 yields ±1.96

Hypothesis Testing t-Tests

- One-Sample t-Test
  - Determine whether the **mean of a single sample** differs significantly from a known/hypothesized population mean

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}}$$

- Two-Sample t-Test
  - Compare the **means of two independent samples** to determine if they differ significantly

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{s_p^2(\frac{1}{n_1} + \frac{1}{n_2})}} \text{ where, } s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$$

Hypothesis Testing Tests

- Paired t-Test
  - Used to compare the **means of two related samples** to determine if they differ significantly
  - $\bar{d}$ is the mean of the differences between paired observations

$$t = \frac{\bar{d}}{s_d/\sqrt{n}}$$

- Chi-Square Tests
  - Determine if there is **a significant association** between categorical variables
  - O_i and E_i represents the observed and expected frequency in each category

$$\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i}$$

ANOVA

- Analysis of Variance (ANOVA) compares the means of three or more groups to determine if at least one of the group means is significantly different from the others
  - Helps testing the hypothesis that all group means are equal against the alternative hypothesis that at least one mean is different
  - k groups, and n_i observations, $N=n_1+n_2+\ldots+n_k$
- **Total Sum of Squares (SST)** measures the total variation in the data

  $$SST=\sum_{i=1}^{k}\sum_{j=1}^{n_i}(X_{ij}-\bar{X})^2$$

- where
  - X_{ij} is the j-th observation in the i-th group
  - $\bar{X}$ is the overall mean of all observations

Data-Driven Decision Making

Random Sampling	Stratified Sampling	Cluster Sampling	Statistical Tests
▪ Each member of the population has an equal chance of being selected	▪ Division of population to subgroups with similar characteristics, sampling	▪ Division of population into clusters, randomly select some clusters, sample all members	▪ Tests to determine if there is enough evidence to reject a null hypothesis

Maximum Likelihood Estimation (MLE)

- Maximum Likelihood Estimation (MLE) is a method for estimating the parameters of a probability distribution by maximizing the likelihood function
- The likelihood is the joint probability of the observed data given the parameters
- Let X_1, X_2, ..., X_n be i.i.d. random variables with PDF $f(x;\Theta)$, where Θ is the parameter to be estimated

  $L(\Theta) = \prod_{i=1}^{n} f(x_i;\Theta)$ (likelihood function)

  $\ell(\Theta) = \log L(\Theta) = \sum_{i=1}^{n} \log f(x_i;\Theta)$ (log-likelihood function)
- The MLE is obtained by solving, i.e. optimization

$$\frac{\partial \ell(\Theta)}{\partial \Theta} = 0$$

Bayesian Estimation

- Bayesian estimation incorporates prior knowledge about the parameter and updates this knowledge based on observed data using Bayes theorem
- Let X_1, X_2, ..., X_n be a sample from a distribution with parameter Θ
- The posterior distribution of Θ given the data
  - $P(\Theta \mid X) = P(X \mid \Theta)\, P(\Theta) \,/\, P(X)$
- where
  - $P(X \mid \Theta)$ is the likelihood function
  - $P(\Theta)$ is the prior
  - and $P(X)$ is the marginal likelihood
- Application Θ: class, X: data points will make the estimation Naive Bayes classifier

Linear Regression

- Models the **relationship between a dependent variable** Y and one or more **independent variables** X by fitting a linear equation to the observed data
- The linear regression model

$$Y = \beta_0 + \beta_1 X + \varepsilon$$

- where
  - β_0 is the intercept term, $\beta_0 = \frac{\sum y - \beta_1 \sum x}{N}$
  - β_1 is the slope, $\beta_1 = \frac{N \sum xy - \sum x \sum y}{N \sum x^2 - (\sum x)^2}$
  - ε is the error term
  - coefficients are solved using the least squares optimization

Polynomial Regression

- Form of a linear regression where the relationship between X and Y is modeled as an **n-th degree polynomial**
- The polynomial regression model

$$Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \ldots + \beta_n X^n + \varepsilon$$

- A polynomial regression is useful to model non-linear relationships, such as when a curve is a better fit rather than a straight line
- A lower-degree polynomial in polynomial regression can help avoid overfitting
- Provides flexibility by allowing various degrees of the independent variables
  - Underlying relationship is complex but avoids a specific higher-order relationship

Logistic Regression

- Logistic regression is used for **modeling binary outcome** variables
- Models the probability that Y belongs to a particular category based X
- The logistic regression model
  - $\mathrm{P}(Y = 1|X) = \frac{1}{1+e^{-(\beta_0+\beta_1 X)}}$
- The log-odds (logit)
  - $\log\left(\frac{\mathrm{P}(Y = 1|X)}{1-\mathrm{P}(Y = 1|X)}\right) = \beta_0 + \beta_1 X$
- One of the best classifiers
- Generally model parameters are fit by gradient descent optimization

Linear Algebra

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Linear Algebra

- Branch of mathematics deals with vector spaces and linear mappings between them
- Study of vectors, vector spaces (also known as linear spaces), linear transformations, and systems of linear equations
- Key concepts include matrices, determinants, eigenvalues, and eigenvectors, which are used to solve a variety of mathematical and real-world problems in engineering, physics, computer science, economics, and statistics
- Foundational tools for understanding and working with multi-dimensional spaces
- $\mathcal{V}$ is a vector space over a field $\mathbb{F}$ if for all vectors $u, v \in \mathcal{V}$ and scalars $\alpha, \beta \in \mathbb{F}$, the operations of vector addition and scalar multiplication satisfy $\alpha u + \beta v \in \mathcal{V}$

Linear Algebra Basics

- **Vector** A vector is an ordered list of numbers
  - represented as column vectors or row vectors

$$\mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix}$$

- **Matrix** A rectangular array of numbers arranged in n rows and m columns

$$\mathbf{A}_{n\times m} = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1m} \\ a_{21} & a_{22} & \dots & a_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nm} \end{bmatrix}$$

Linear Algebra Basics

- **Matrix Multiplication**
  - The product of $\mathbf{A}_{n\times m}$ and $\mathbf{B}_{m\times p}$ is a matrix $\mathbf{C}_{n\times p}$

$$\mathbf{C} = \mathbf{AB} \text{ where } c_{ij} = \sum_{k=1}^{m} a_{ik} b_{kj}$$

- **Identity Matrix**
  - A square matrix with ones on the diagonal and zeros elsewhere
  - Multiplicative identity,

$$\mathbf{AI} = \mathbf{IA} = \mathbf{A}$$

$$\mathbf{I}_{n\times n} = \begin{bmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{bmatrix}$$

Linear Algebra Basics

- **Determinant**
  - The determinant of a square matrix $\mathbf{A}$ is a scalar value that is a function of the entries of $\mathbf{A}$

$$\det(\mathbf{A}) = \sum_{i=1}^{n} (-1)^{i+1} a_{1i} \det(\mathbf{A}_{1i})$$

- The determinant provides several characteristic information about the matrix such as existence of inverse, eigenvalues and characteristic polynomial, and measure of linear dependence

- **Inverse Matrix**
  - The inverse of a square matrix $\mathbf{A}$ is a matrix $\mathbf{A}^{-1}$ such that

$$\mathbf{A}\mathbf{A}^{-1} = \mathbf{I}$$

$$\mathbf{A}^{-1} = \frac{1}{\det(\mathbf{A})} \operatorname{adj}(\mathbf{A})$$

Linear Algebra

- **Dot Product**
  - The dot product of two vector u and v is a scalar

$$\mathbf{u} \cdot \mathbf{v} = \sum_{i=1}^{n} u_i v_i$$

- Geometric meaning can be interpreted as the product of the magnitude of u and the magnitude of the projection of v onto u
  - represents how much of v lies in the direction of u

- **Matrix Vector Multiplication**
  - The product of a matrix A and a vector v is a vector w

$$\mathbf{w} = \mathbf{A}\mathbf{v} \text{ where } w_i \sum_{j=1}^{n} a_{ij} v_j$$

Linear Algebra Eigenvalues

- **Eigenvalues and Eigenvectors**
- A square matrix $\mathbf{A}$, if there is a scalar λ and a non-zero vector $\mathbf{v}$ such that $\mathbf{Av} = \lambda\mathbf{v}$, then λ is an eigenvalue and $\mathbf{v}$ is an eigenvector of $\mathbf{A}$
  - SVD can solve a non-square $\mathbf{A}$

```
from sklearn.decomposition
  import PCA
Xs = StandardScaler(
  ).fit_transform(X)
# Visualize the models above in 2D
Xpca = PCA(n_components=2
  ).fit_transform(Xs)
```

- Systems Stability Analysis
  - Negative eigenvalues of a system's matrix indicate stability
- Principal Component Analysis
  - Dimensionality reduction
- Graph Theory
  - Eigenvalues of the adjacency matrix indicate connectivity
- Image Processing
  - Eigenfaces use eigenvectors to capture the essential features

A Note on Data Structures

- Often require shaping tensors using the `numpy.reshape()` or `torch.view()`
- Python, PyTorch, Numpy, C++ languages have row-major order
- Matlab and Fortran have column-order

```
# row-major order
A = np.array([[1,2,3,4],[5,6,7,8]])
print(A)
print(A.reshape(4,-1))
print(A.reshape(-1,2))

# output
[[1 2 3 4] [5 6 7 8]]
[[1 2] [3 4] [5 6] [7 8]]
[[1 2] [3 4] [5 6] [7 8]]
```

Classical Generative Algorithms

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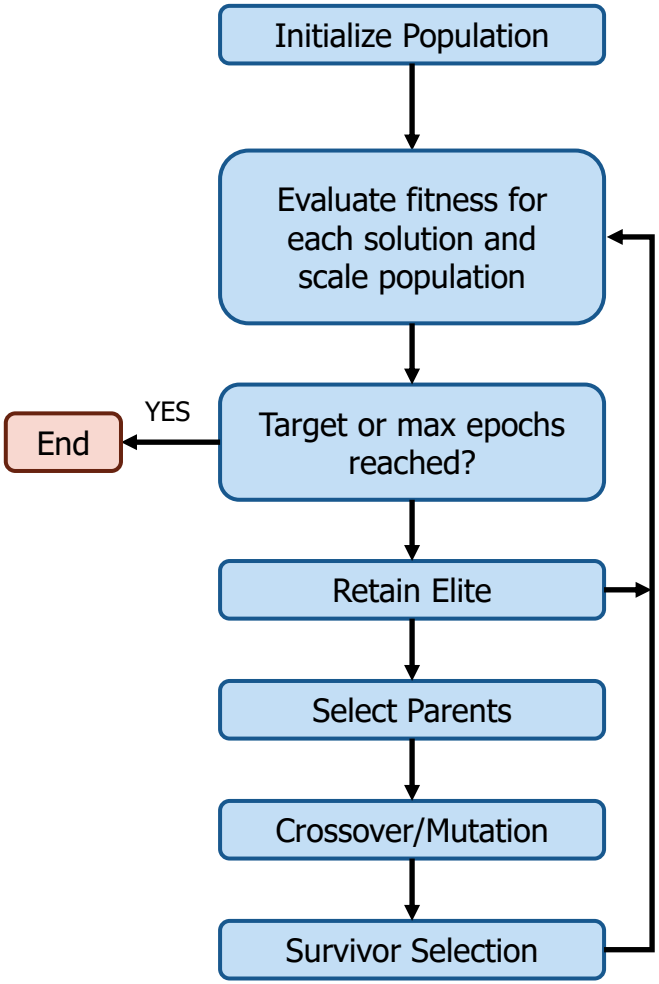
Genetic Algorithms

- GAs mimic natural selection for **optimization**
- Solve constrained and unconstrained problems by evolving solutions over generations
- GAs are effective for complex, non-traditional optimization challenges
- Well-suited for mixed data structures (e.g., integers, strings, binary)
- Solutions improve through mutation, selection, and recombination



Genetic Algorithm Primary Components

- **Population** Bit strings, real, integer, binary, any data structure that represents a solution

- **Fitness/Objective** Function or objective you want to optimize for

- **Retain Elite** Keep best performing solutions for the next generation, to maintain optimal solutions for each generation

- **Crossover**

P1 (01101000) → (01101010) C1

P2 (11011010) → (11011000) C2

- **Mutation**

Before (01101010)

After (01111010)

- **Survivor Selection** Ensure that the **fitter** solutions and diversity are maintained in the population

Genetic Algorithm Primary Components

- **Fitness Function** The fitness function $\mathrm{f}(x)$ evaluates how a given solution x is close to the optimum
  - The fitness value determines the probability of selection for reproduction
- **Selection Probability** The selection probability P_i for an individual i is given by

$$\mathrm{P}_i = \frac{\mathrm{f}(x_i)}{\sum_{j=1}^{N} \mathrm{f}(x_j)}$$

  - where N is the population size and $\mathrm{f}(x_j)$ is the fitness of individual j

GA Three Types of Rules

- **Selection Rules** Used to select individual solutions that contribute to the population during the next generation
  - Selection occurs stochastically and can depend on the individual's fitness/score
- **Crossover Rules** Recombines two or more solutions to form hybrid solutions for the next generation
- **Mutation Rules** Perturbs and apply random changes to individual solutions to explore different possible solutions

GA Pseudocode

Input: Population size N, Fitness function F, Crossover rate r_c, Mutation rate r_m, Number of generations G

Output: Optimized solution from the final population

1. **Initialize** population with N random solutions
2. **For** generation $g = 1$ to G do
   - **Evaluate** fitness of each individual using F
   - **Select** individuals for reproduction based on fitness
   - **Perform** crossover on selected individuals at rate r_c to create offspring
   - **Apply** mutation to offspring at rate r_m
   - **Replace** least fit individuals in the population with offspring
3. **End for**
4. **Return** the best solution from the final population

GA Example Maximize a Quadratic Func

- **Problem Statement** Maximize the function $f(x) = -(x-2)^2 + 4$ which has a maximum value of 4 at $x = 2$
- **Initialization** Generate an initial population of 4 candidate solutions
  - $\{x_1, x_2, x_3, x_4\}$
- **Fitness Evaluation** Evaluate the fitness of each candidate $f(x) = -(x-2)^2 + 4$
- Assume the initial population is $\{0, 1, 3, 4\}$
  - $f(0) = -(0-2)^2 + 4 = 0$
  - $f(1) = -(1-2)^2 + 4 = 3$
  - $f(3) = -(3-2)^2 + 4 = 3$
  - $f(4) = -(4-2)^2 + 4 = 0$

Example (cont)

- **Selection** Calculate the selection probabilities
  - $P(0) = \frac{0}{0+3+3+0} = 0$
  - $P(1) = \frac{3}{0+3+3+0} = 0.5$
  - $P(3) = \frac{3}{0+3+3+0} = 0.5$
  - $P(4) = \frac{0}{0+3+3+0} = 0$
- **Crossover** Perform crossover to generate new offspring
  - Suppose single-point crossover is applied at the midpoint for $x_1 = 1$ and $x_2 = 3$
- **Mutation** Apply mutation with a small probability (e.g., 0.1)
  - Assume no mutation occurs in this iteration

Example (cont)

- **Evaluation and Iteration** Evaluate the fitness of the new population and repeat the selection, crossover, and mutation steps until convergence
- Next Generation Example
  - Assume the next gen population after selection andcrossover is $\{1.5,\ 2.5,\ 3,\ 4\}$
  - $f(1.5) = -(1.5-2)^2 + 4 = 3.75$
  - $f(2.5) = -(2.5-2)^2 + 4 = 3.75$
  - $f(3) = -(3-2)^2 + 4 = 3$
  - $f(4) = -(4-2)^2 + 4 = 0$
- Update the selection probabilities and iterate further until convergence to the optimal solution

GAs and Generative AI

- **Optimization** GAs optimize complex problems by iteratively evolving solutions, applicable in tuning Generative AI models or optimizing neural network architectures
- **Model Evolution** In Generative AI, particularly in neural network training, GAs can evolve models by selecting and combining high-performing models or hyperparameters, similar to how GAs evolve solutions in other optimization tasks
- **Creative Exploration** GAs can be used to generate diverse data samples, which can be input for Generative AI models
- **Data Generation** GAs generate synthetic data for training Generative AI models
- **Hybrid Approaches** Some advanced Generative AI models combine GAs with other machine learning techniques to explore and generate better solutions
  - For example, GAs might be used to optimize the parameters of a GAN

Time Series

- In classical ML, a single data point used $x \in \mathbb{R}^M$ to model and predict
  - Assumption of M many independent features
  - Assumption of data points have order independence
    - A shuffle would not change the performance
- The knowledge can be carried by a sequence of data points $\langle x_t \rangle$
  - Speech, video, events on a network
  - Then the learning method is called sequence learning
- A sequence to sequence model, $\mathbf{v}$ is encoded input vector

$$\mathrm{P}(y_1 \dots y_{T0} \mid x_1 \dots x_T) = \prod_{t=1}^{T0} \mathrm{P}(y_t \mid \mathbf{v}, y_1 \dots y_{t-1})$$

Markov Chains

- A sequence of random variables $X_1, X_2, \ldots$, where each variable X_i represents the state of the system at step i
- The system transitions between these states in a discrete space $S = \{s_1, s_2, \ldots, s_n\}$
- The probability of moving from one state to another is described by the transition probability matrix $P \in \mathbb{R}^{n \times n}$
  - where each element P_{ij} represents the probability of transitioning from state s_i to state s_j with $P_{ij} = \Pr(X_{t+1} = s_j \mid X_t = s_i)$
- $X_t = s_i$ is called the conditioning event for the conditional probability
- The sum of the probabilities of transitioning from a state s_i to any other state must be 1 (Why?) i.e., $\sum_j P_{ij} = 1$

Sequence Learning

- The goal is to learn the transition probabilities from a given set of observed sequences
- Given a set of training sequences, we can estimate the transition probability P_{ij} as the relative frequency of transitions from state s_i to state s_j

$$P_{ij} \approx \frac{\text{Number of transitions from } s_i \text{ to } s_j}{\text{Total number of transitions from } s_i}$$

- These transition probabilities can be represented in a matrix, which is then used to describe the dynamics of the system

Sequence Generation

- Once the model has been trained and the transition probability matrix P has been learned, sequences can be generated by starting from an initial state s_0 and making transitions according to the learned probabilities
- The generation of a sequence is done by following these steps:
  - Select an initial state s_0 from the state space S
  - Generate the next state s_1 by sampling from the probability distribution $P(s_0)$
  - Repeat this process for the desired number of steps
- Or formally written as, $s_{t+1} \sim P(s_t)$ where s_{t+1} is the state at time $t+1$ and $P(s_t)$ is the probability distribution over the next states given the current state s_t

Higher-Order Markov Models

- First-order Markov models assume that the next state depends only on the current state
- Higher-order models relax this assumption by allowing the next state to depend on a fixed number of previous states
- For an n-th order Markov model, the probability of transitioning to a new state depends on the previous n states,

$$\Pr(X_{t+1} = s_j \mid X_t = s_i, X_{t-1} = s_{i-1}, \dots, X_{t-n+1} = s_{i-n+1})$$

- While higher-order Markov models can capture more complex dependencies in the data, they also require significantly more parameters, making them more difficult to estimate accurately from data

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